

# Software Requirements Engineering

## Assignment #1

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### **Case Study:**

Gerhard, a senior manager at Contoso Pharmaceuticals, was meeting with Cynthia, the manager of Contoso's IT department. "We need to build a chemical tracking information system," Gerhard began. "The system should keep track of all the chemical containers we already have in the stockroom and in laboratories. That way, the chemists can get some chemicals from someone down the hall instead of always buying a new container. This should save us a lot of money. Also, the Health and Safety Department needs to generate government reports on chemical usage and disposal with a lot less work than it takes them today. Can you build this system in time for the compliance audit in five months?"

## **1. Business Requirements**

### **1.1 Background**

Contoso Pharmaceuticals currently lacks an efficient centralized system for tracking chemical containers stored in stockrooms and laboratories. As a result, chemists may purchase new chemical containers even when the same chemicals are already available elsewhere within the organization. This leads to unnecessary procurement costs, inefficient resource utilization, and poor visibility of chemical inventory.

In addition, the Health and Safety Department faces difficulty in preparing government compliance reports related to chemical usage and disposal. The current manual or fragmented process is time-consuming, labor-intensive, and prone to error. With an upcoming compliance audit in five months, Contoso Pharmaceuticals requires an automated and reliable chemical tracking information system to improve operational efficiency and support regulatory compliance.

### **1.2 Business Opportunity**

The proposed Chemical Tracking Information System presents Contoso Pharmaceuticals with an opportunity to reduce redundant chemical purchases by improving visibility into existing stock across stockrooms and laboratories. Chemists will be able to identify available chemicals within the organization before requesting new orders.

The system will also streamline the preparation of mandatory government reports on chemical usage and disposal, reducing administrative effort and improving reporting accuracy. This creates an opportunity for better compliance management, cost savings, improved workplace safety, and more effective inventory control.

## 1.3 Business Objectives

The following business objectives are identified for the proposed system:

1. Reduce unnecessary chemical purchases by enabling employees to locate available chemicals already present in stockrooms or laboratories.
2. Improve visibility and accuracy of chemical container records across the organization.
3. Enable the Health and Safety Department to generate government compliance reports more quickly and with less manual effort.
4. Improve tracking of chemical usage, movement, and disposal to support safety and regulatory requirements.
5. Deliver the system in time to support the compliance audit scheduled in five months.
6. Increase operational efficiency in chemical inventory management and reporting processes.

### Business Objectives Model

**Goal:** Improve chemical inventory control and regulatory compliance at Contoso Pharmaceuticals

- **Objective 1:** Minimize duplicate chemical purchases
  - Provide visibility into chemical availability
  - Allow chemists to search for chemicals across locations
- **Objective 2:** Improve compliance reporting
  - Record chemical usage and disposal data accurately
  - Automate generation of government reports
- **Objective 3:** Improve operational efficiency
  - Centralize tracking of chemical containers

- Reduce manual paperwork and administrative work
- **Objective 4:** Meet audit readiness deadline
  - Deploy initial system within five months
  - Ensure required reporting functionality is available before audit

## 1.4 Vision Statement

For Contoso Pharmaceuticals, which needs better control over chemical inventory and compliance reporting, the Chemical Tracking Information System is an internal information management solution that provides centralized tracking of chemical containers, their locations, usage, and disposal. Unlike the current manual and fragmented approach, the proposed system improves inventory visibility, reduces duplicate purchases, supports regulatory compliance, and increases efficiency in chemical management processes.

## 1.5 Business Risks

Several risks may affect the success of the project:

1. The system may not be completed within the five-month deadline required for the compliance audit.
2. Existing chemical inventory records may be incomplete, inconsistent, or inaccurate, affecting initial data entry.
3. Users such as chemists and laboratory staff may resist adopting the new system if it changes their routine workflows.
4. Failure to capture all required compliance data could reduce the usefulness of government reports.
5. Integration challenges may arise if the new system must interact with existing procurement or safety management processes.
6. Inadequate training may result in incorrect data entry or underuse of the system.
7. Changes in government reporting requirements may require modifications to the system after development.

## 1.6 Business Assumptions and Dependencies

### Assumptions

1. Contoso Pharmaceuticals is willing to allocate sufficient budget, staff, and technical resources for the system development.

2. Laboratory and stockroom personnel will cooperate in providing current inventory information.
3. The reporting requirements for government compliance are known and can be incorporated into the system.
4. Users will be trained before the system goes live.
5. The required infrastructure, such as computers and network access, is already available in laboratories and stockrooms.

## **Dependencies**

1. The success of the system depends on the availability and accuracy of existing chemical inventory data.
2. The project depends on timely input from the Health and Safety Department regarding compliance reporting requirements.
3. The system depends on active participation by chemists, laboratory staff, and stockroom staff.
4. The project timeline depends on approval from senior management and support from the IT department.
5. Audit readiness depends on successful delivery of the initial release before the compliance audit date.

# **2. Scope and Limitations**

## **2.1 Major Features**

The Chemical Tracking Information System will include the following major features:

1. Chemical container registration and inventory management
2. Search and lookup of chemicals by type, quantity, and location
3. Tracking of chemical movement between stockrooms and laboratories
4. Recording of chemical usage and disposal
5. Compliance reporting for government requirements
6. User access and role management
7. Alerts or notifications for low stock, expired chemicals, or disposal needs

## **Feature Tree**

### **Chemical Tracking Information System**

- **Inventory Management**

- Add new chemical container
  - Update container details
  - View stock levels
  - Record container status
- **Chemical Search**
  - Search by chemical name
  - Search by location
  - Search by quantity/availability
- **Location Tracking**
  - Track stockroom containers
  - Track laboratory containers
  - Transfer container between locations
- **Usage and Disposal Tracking**
  - Record chemical usage
  - Record disposal details
  - Maintain disposal history
- **Compliance Reporting**
  - Generate usage reports
  - Generate disposal reports
  - Prepare government compliance reports
- **User Management**
  - User login
  - Role-based access
  - Audit trail of user actions
- **Notifications**
  - Low stock alert
  - Expiry alert
  - Disposal reminder

## **2.2 Scope of Initial and Subsequent Releases**

### **Initial Release**

The initial release will focus on the most critical features required for immediate business value and audit readiness, including:

1. Registration of chemical containers
2. Centralized inventory database for stockrooms and laboratories
3. Search functionality to locate available chemicals
4. Recording of chemical usage and disposal
5. Generation of essential compliance reports

6. Basic user authentication and access control

This release must be completed within five months to support the compliance audit.

## **Subsequent Releases**

Later releases may include the following enhancements:

1. Advanced analytics and dashboards for chemical usage trends
2. Automated alerts for expired or low-stock chemicals
3. Integration with procurement systems to automate reordering
4. Barcode or RFID support for faster chemical tracking
5. Mobile or tablet-based access for laboratory staff
6. Expanded audit trail and advanced security controls

## **2.3 Limitations**

The following limitations apply to the proposed system:

1. The system will track chemical containers within Contoso Pharmaceuticals only; it will not manage supplier or external distribution activities.
2. The initial release may not include advanced automation features such as barcode scanning or procurement integration.
3. The system depends on users entering and updating data accurately; incorrect input will affect system reliability.
4. The system will support compliance reporting only for the government requirements identified during development.
5. Real-time tracking may be limited if users fail to promptly record movement, usage, or disposal of chemicals.
6. The system is intended primarily for internal management and compliance purposes, not for full laboratory process automation.